Project #49: GrammarForge

AI Portfolio Project Report

Project Overview

Project: #49 - GrammarForge

Language: OCaml 5.3.0

Framework: Dream 1.0.0~alpha8

Lines of Code: 1057

Date: 2026-03-29

GitHub: https://github.com/malbinjvc/grammar-forge

GrammarForge is an AI-powered grammar and syntax analysis REST API built with OCaml and the Dream web framework. It provides text analysis for readability, grammar checking, sentence structure analysis, and AI-powered writing suggestions via a mock Claude client. Features async handlers using lwt_ppx syntax.

Architecture

- Dream web framework with async Lwt handlers
- lwt_ppx preprocessor for let%lwt async syntax
- In-memory data store for text documents
- Service layer in lib/services.ml for analysis logic
- Client layer in lib/clients.ml for AI integration
- Type definitions in lib/types.ml with Yojson serialization
- Route definitions in lib/routes.ml

Key Features

- Text readability analysis (Flesch-Kincaid grade level)
- Grammar checking with rule-based error detection
- Sentence structure analysis (simple, compound, complex)
- Passive voice and adverb detection
- AI-powered writing suggestions via mock Claude client
- Word frequency analysis and vocabulary metrics
- Bulk text comparison and scoring

API Endpoints

- GET /health - Health check
- POST /api/texts - Submit text for analysis
- GET /api/texts - List all submitted texts
- GET /api/texts/:id - Get text by ID
- POST /api/texts/:id/analyze - Full grammar analysis
- POST /api/texts/:id/readability - Readability score
- POST /api/texts/:id/suggestions - AI writing suggestions
- POST /api/compare - Compare two texts

Testing

Test Results: 42/42 passing

Errors Encountered: 9

Errors Resolved: 9

- dune allow_empty error for package definition
- Unused rec flag on non-recursive function
- Missing lwt_ppx preprocessor for let%lwt syntax
- libev not found on macOS (Homebrew non-standard path)
- String.contains_s does not exist in OCaml stdlib
- Passive voice/adverb detection failing due to trailing punctuation
- CI OCaml version mismatch (5.2 vs 5.3 requirement)
- Docker base image OCaml version mismatch
- Docker build binary not produced due to incomplete deps

Docker

- Multi-stage build with ocaml/opam:debian-12-ocaml-5.3
- opam install . --deps-only for proper dependency resolution
- Explicit lwt_ppx installation and dune build bin/main.exe
- debian:bookworm-slim runtime
- Non-root user (grammarforge)
- Healthcheck via curl against /health

CI/CD Pipeline

- GitHub Actions with ocaml/setup-ocaml@v3 (OCaml 5.3)
- opam install --deps-only --with-test
- opam exec -- dune build and dune test
- Docker build verification
- All jobs passing

Final Status

Build: Passing

Tests: 42/42 green

CI: All green

Docker: Build successful

GitHub: Pushed and live